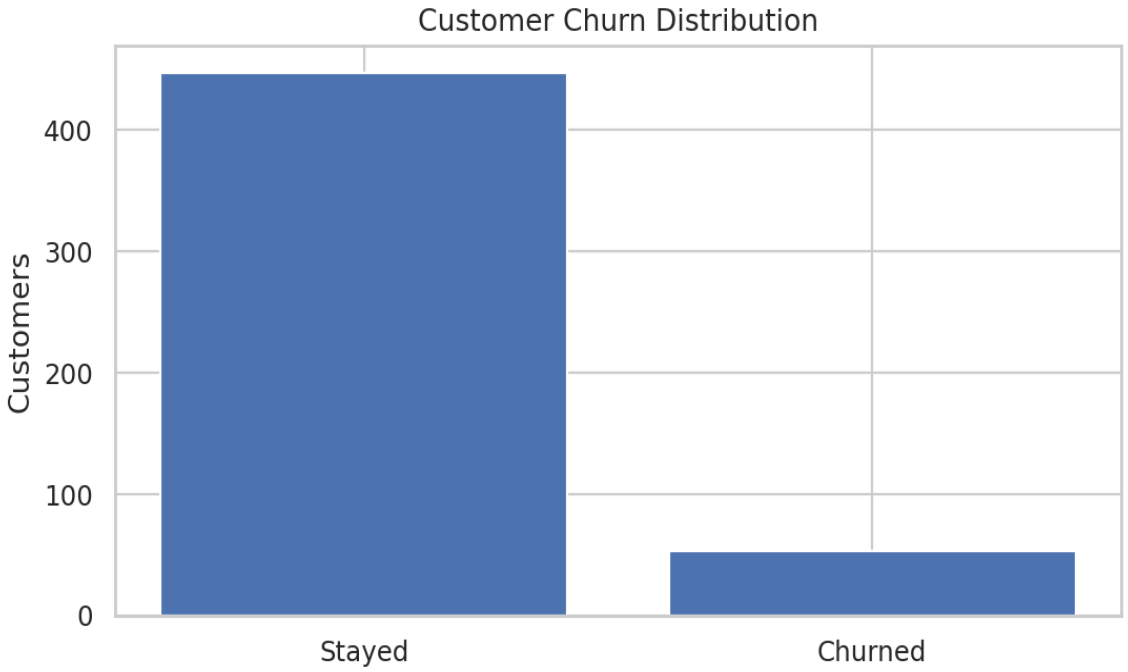
Technical Report

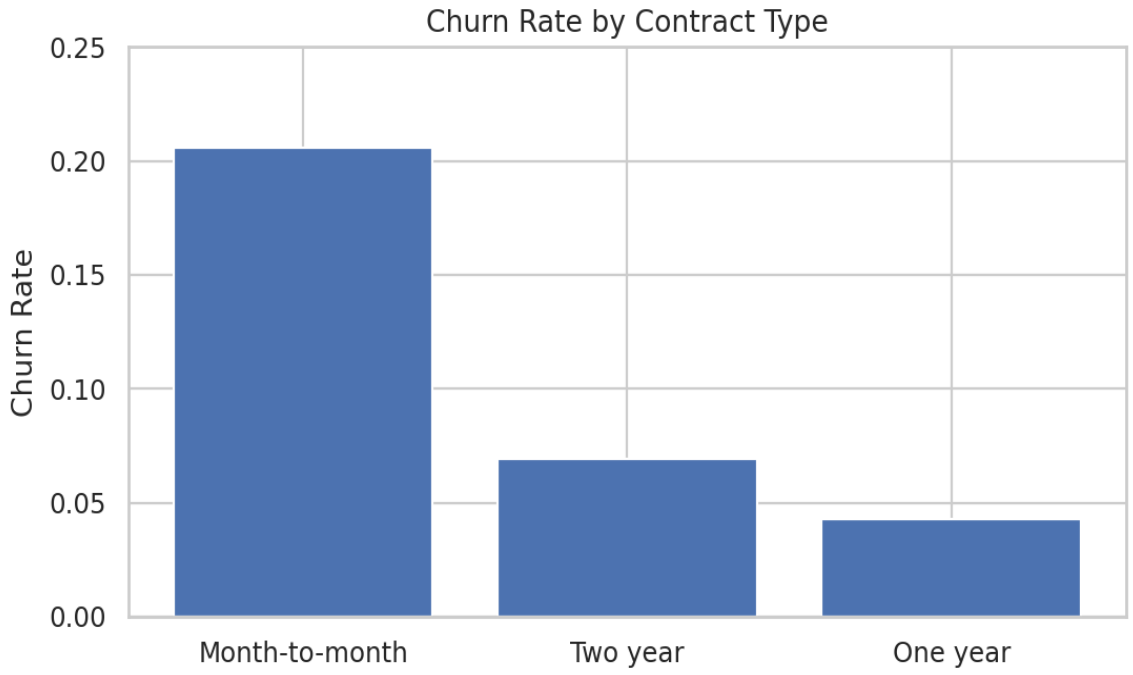
1. Business Context

Customer churn threatens recurring revenue. The project follows an end-to-end workflow from business framing to implementation.



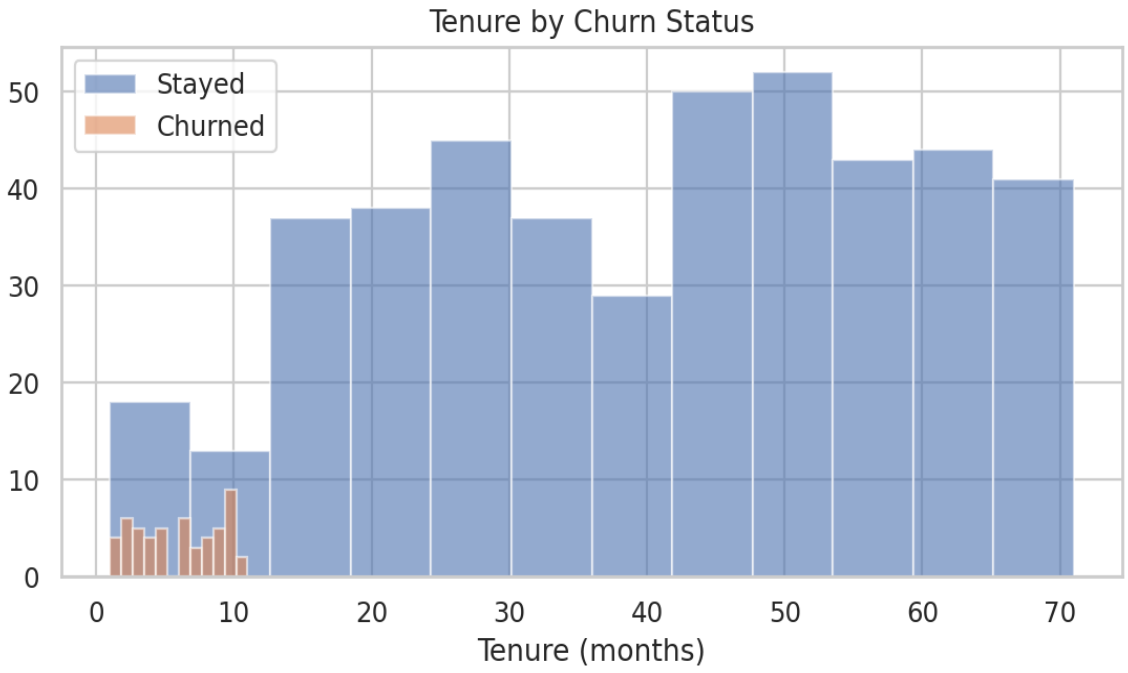
2. Data Quality

500 rows, 9 original columns, zero missing values and zero duplicate rows. Raw and cleaned files are separated.



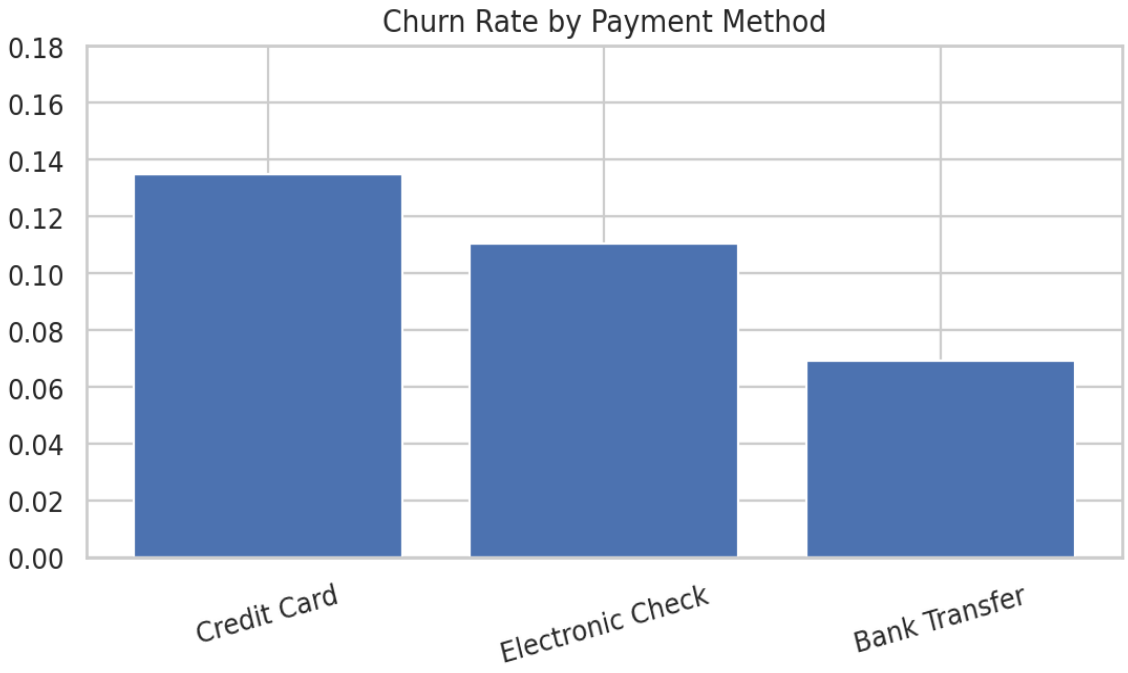
3. Exploratory Analysis

Contract and tenure segments show meaningful churn differences. Overall churn is 10.6%.



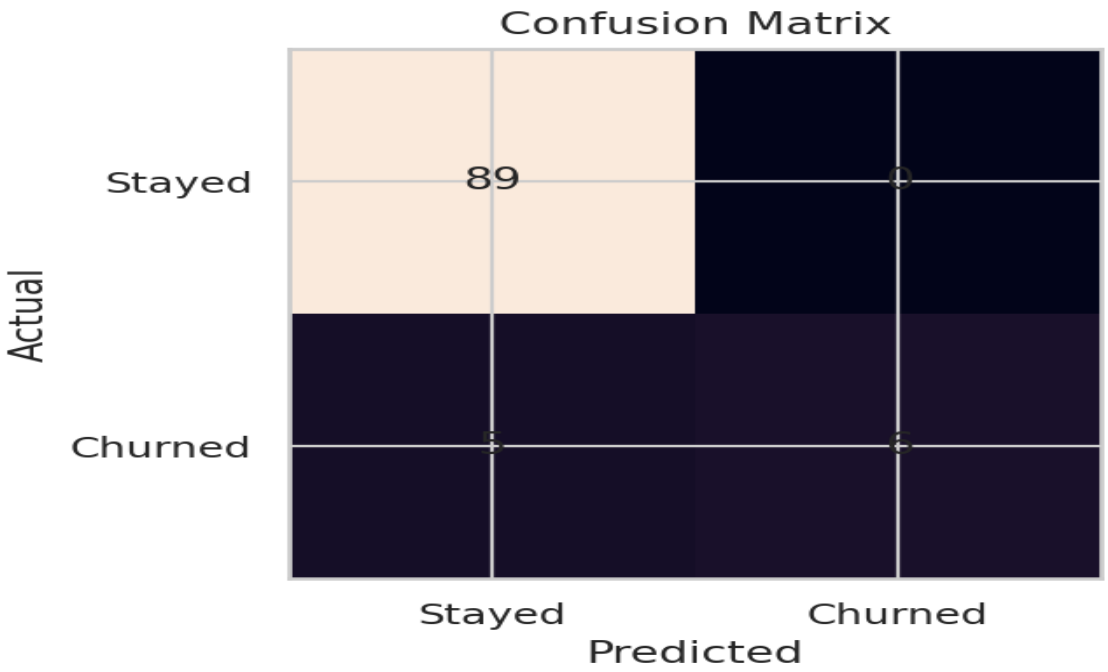
4. Statistical Validation

Contract chi-square p=9.59e-07; tenure Welch t-test p=8.98e-130; monthly-charge Welch t-test p=0.01008.



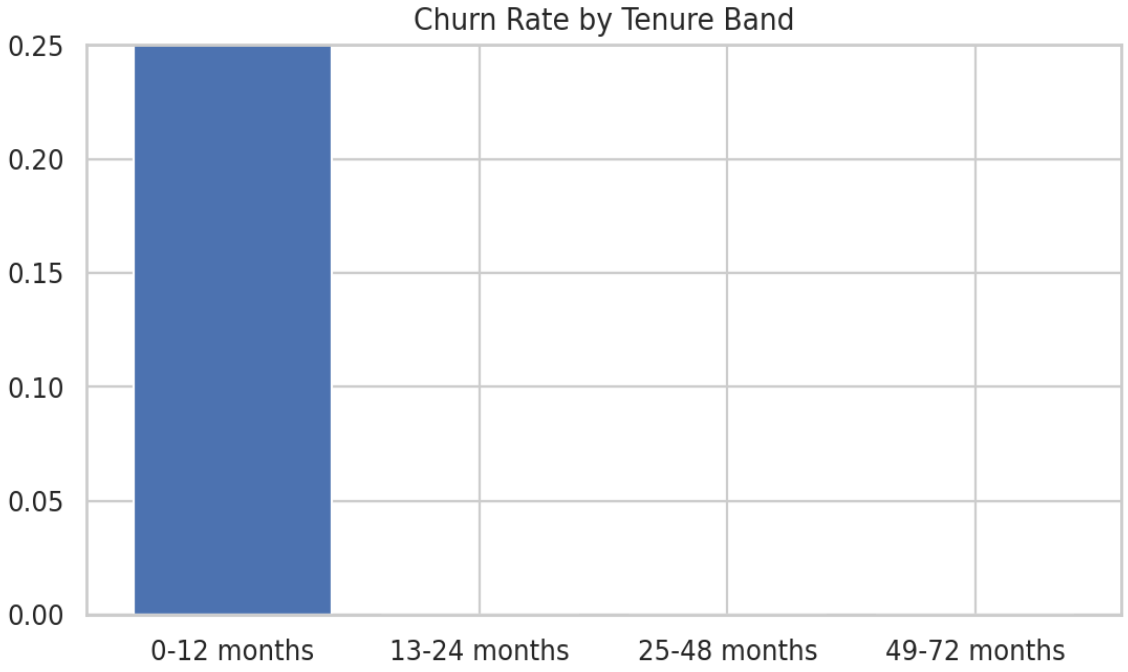
5. Predictive Modeling

Logistic regression: accuracy 95.0%, precision 100.0%, recall 54.5%, F1 70.6%, ROC-AUC 0.981.



6. Recommendations

Use targeted contract conversion, early-tenure support and payment assistance experiments. Use the model for prioritization, not automatic punitive decisions.



7. Implementation & Reproducibility

0–30 days baseline/pilot; 31–60 days controlled experiments; 61–90 days scale winners. Fixed seed, structured notebooks, requirements and reusable script support reproducibility.

